Extrapolating the biodegradability of 3D-Printing polymers based on their tensile strength

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Background

3D-Printing is an increasingly popular tool to manufacture physical parts and components in several different material types, like plastics, metal and ceramics. The interest of this experiment lies in the biodegradability of plastics/polymers like PLA, TPU & ABS. Polymers used in 3D-Printing are considered biodegradable, but only in industry composting and anaerobic digestion. My goal with this experiment is to predict, using polymer tensile strength properties, and deduce what filament would endure the longest before any observable, natural degradation.

Problem and Significance

3D-Printing filaments are only biodegradable at industrial conditions, as opposed to natural environments and everyday use. This drastically amplifies their impact on the global plastic footprint. The significance of this experiment is to correlate biodegradability and tensile strength, to extrapolate the tendency of polymers to be made 'greener', and provide a knowledge base for further study analysing the time taken for degradability.

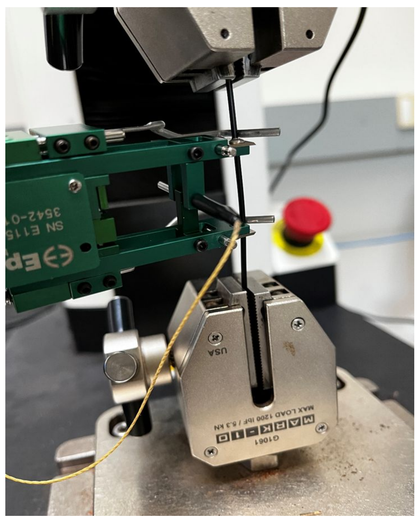


Figure 1: PLA Filament displaced on a tensile tester, attached with an extensometer.

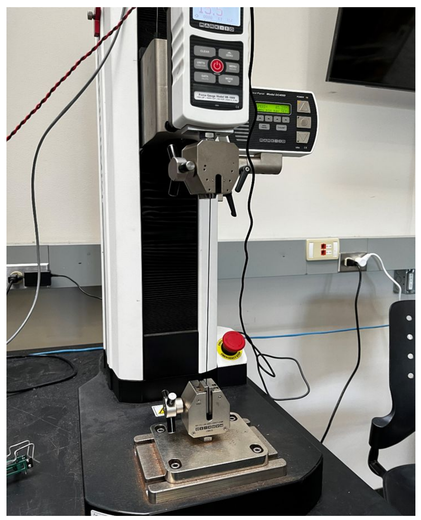


Figure 2: TPU Filament displaced on a tensile tester

Impact

Achieving a faster biodegradation processes in natural environment can reduce problems of excessive plastic waste, save finances on waste management, and contribute to a healthier ecosystem globally. Some of the unanswered questions for this topic of research is to find a chemically-efficient way to accelerate the rate of decomposition for 3D-Printing filaments. A larger scope for this project would be to test the decomposition of these polymers, and superposition the test results onto their tensile strength.

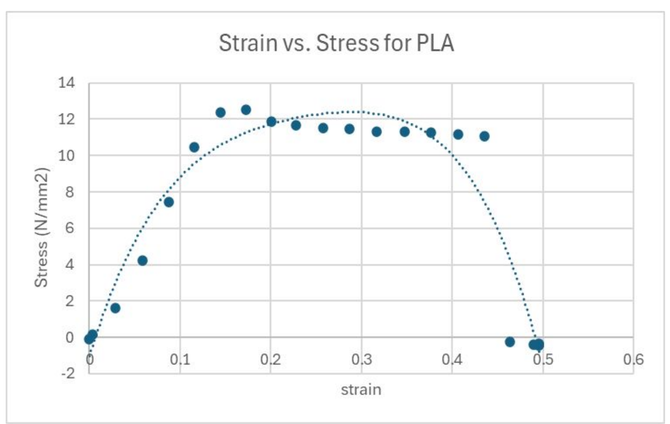


Figure 3: Stress-Strain Curve for PLA

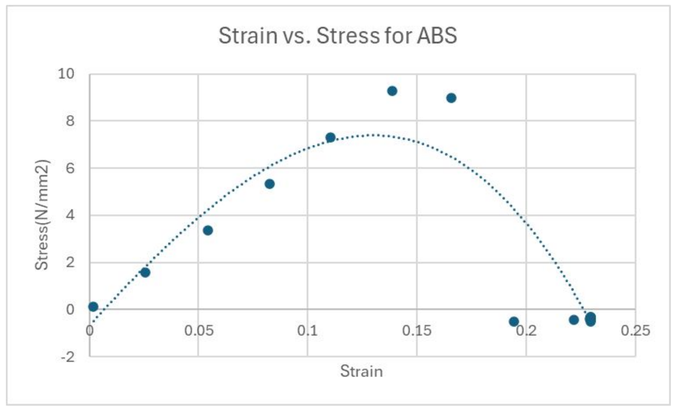


Figure 4: Stress-Strain Curve for ABS

Conclusion

There were several extraneous variables encountered as this experiment was conducted, including, moisture level, length of filament, ambient temperature. It was found that TPU can endure linearly increasing forces for ~180 seconds before the clamps break the ends of the filament (Figure 2). The highest Young's Modulus found was for PLA (92.26) as compared to TPU(7.825) and ABS(69.69). My conclusion is that TPU requires less harsher conditions to degrade than PLA. This is conflicting to the research papers referenced below. This will allow us to identify any exceptions which do not follow a linear positive correlation between tensile strength and biodegradability, further introducing factors such as chemical composition, temperature, and moisture levels in the filament.

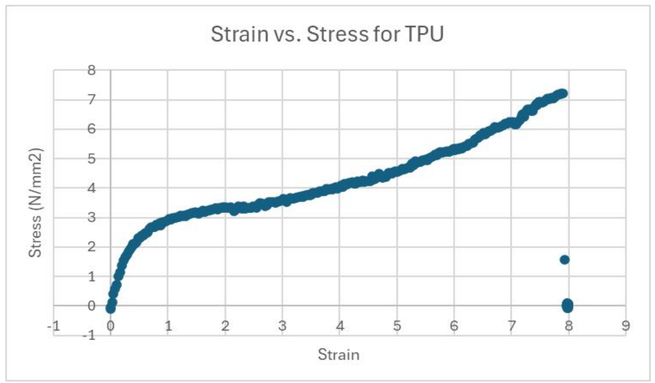


Figure 5: Stress-Strain Curve for TPU

Acknowledgments

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References

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